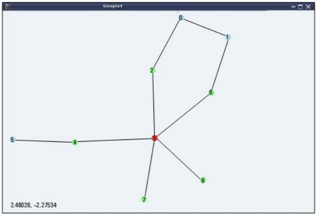

Figure 1: Graph connectivities

```
...
0 errors, 0 warnings
(%i2) g: create_graph(9, [[0, 1], [0, 2], [1, 8], [8, 3], [2, 3],
[3, 4], [4, 5], [3, 6], [3, 7]]);
(%o2)                     GRAPH(9 vertices, 9 edges)
(%i3) vertex_connectivity(g);
(%o3)                                    1
(%i4) edge_connectivity(g);
(%o4)                                   1
(%i5) shortest_path(0, 7, g);
(%o5)                              [0, 2, 3, 7]
(%i6) vertex_distance(0, 7, g);
(%o6)                                   3
(%i7) girth(g);
(%o7)                                   5
(%i8) diameter(g);
(%o8)                                   4
(%i9) radius(g);
(%o9)                                   2
(%i10) graph_center(g);
(%o10)                                 [3]
(%i11) graph_periphery(g);
(%o11)                               [5, 1, 0]
```

Vertex 3 is the only centre of the graph, and 0, 1 and 5 are the peripheral vertices of the graph.

Graph colouring

Graph colouring has been a fascinating topic in graph theory, right since its inception. It is all about colouring the vertices or edges of a graph in such a way that no adjacent elements (vertex or edge) have the same colour.

The smallest number of colours needed to colour the vertices of a graph, such that no two adjacent vertices have the same colour, is called the chromatic number of the graph. *chromatic_number(<g>)* computes the same. *vertex_coloring(<g>)* returns a list representing the colouring of the vertices of <g>, along with the chromatic number.

The smallest number of colours needed to colour the edges of a graph, such that no two adjacent edges have the same colour, is called the chromatic index of the graph. *chromatic_index(<g>)* computes the same. *edge_coloring(<g>)* returns a list representing the colouring of the edges of <g>, along with the chromatic index.

The following demonstration continues colouring the graph from the above demonstration:

```
(%i12) chromatic_number(g);
(%o12)                              3
(%i13) vc: vertex_coloring(g);
(%o13) [3, [[0, 3], [1, 1], [2, 2], [3, 1], [4, 2], [5, 1], [6,
2], [7, 2], [8, 2]]]
(%i14) chromatic_index(g);
(%o14)                              5
(%i15) ec: edge_coloring(g);
(%o15) [5, [[[0, 1], 1], [[0, 2], 2], [[1, 8], 2], [[3, 8], 5],
[[2, 3], 1], [[3, 4], 2], [[4, 5], 1], [[3, 6], 3], [[3, 7], 4]]]
(%i16) draw_graph(g, vertex_coloring=vc, edge_coloring=ec, vertex_
size=5, edge_width=3, show_id=true)$
(%i17) quit();
```

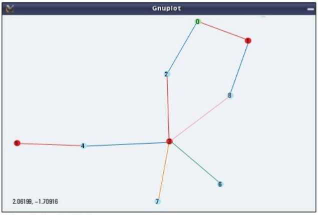

Figure 2: Graph colouring

Figure 2 shows the coloured version of the graph, as obtained by %i16.

Bon voyage

With this article, we have completed a two-year long mathematical odyssey through open source, starting from mathematics in Shell, covering Bench Calculator and Octave, and concluding with Maxima. I take this opportunity to thank my readers and wish them bon voyage with whatever they have gained through our interactions. However, this is not the end—get set for our next journey. END

By: Anil Kumar Pugalia

The author is a gold medallist from NIT Warangal and IISc Bengaluru. Mathematics and knowledge-sharing are two of his many passions. Learn more about him at *http://sysplay.in*. He can be reached at email@sarika-pugs.com.